How Machines Understand Text

Foundations of Natural Language Processing

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Where did you last interact with any technology that understands language?

Type your answer in the chat

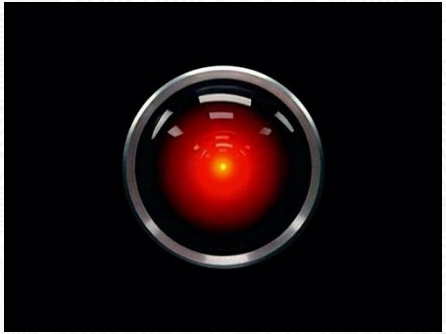


From fiction…

HAL 9000

2001: A Space Odyssey (1968)

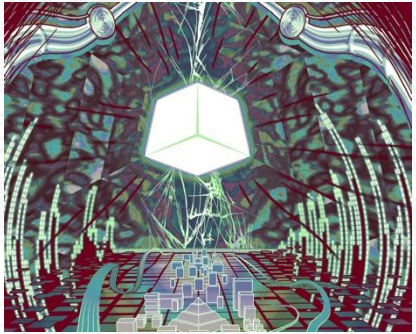
"I'm sorry, Dave. I'm afraid I can't do that."



Wintermute

Neuromancer (novel)

“Nowhere. Everywhere. I’m the sum total of the works, the whole show.”



Ultron

Avengers: Age of Ultron (2015)

"I had strings, but now I'm free. There are no strings on me..."



Wintermute illustration by Kelly Knowles, www.kellyknowles.art

To Reality

Spam filters

Late 20th century

Automated tools that classifies textual data to protect the user.

Virtual Assistants

Siri, Alexa, Google, etc

Can interpret speech to answer questions, follow commands, etc.

LLMs

ChatGPT, Gemini, Claude, etc

Uses a transformer model to replicate language patterns. Chugs water.

Natural Language Processing

- Intersection of computer science, artificial intelligence, and linguistics
- Allow computers to understand, process, and generate human language *meaningfully*
- Two important applications
  - Natural language understanding
  - Natural language generation

Today’s Agenda

1

Text to Tokens

Break text into small pieces that the computer can digest.

2

Numerical Representation

Turning tokens into numbers the computer can understand.

3

Intro to Applications

Sentiment analysis, classification, and topic modeling

Section 1. From Written Text to Tokens

What is a token?

A small unit of text, usually a word or a punctuation mark.

Computers can't read text, but they can count and compare.

“The movie was great!”

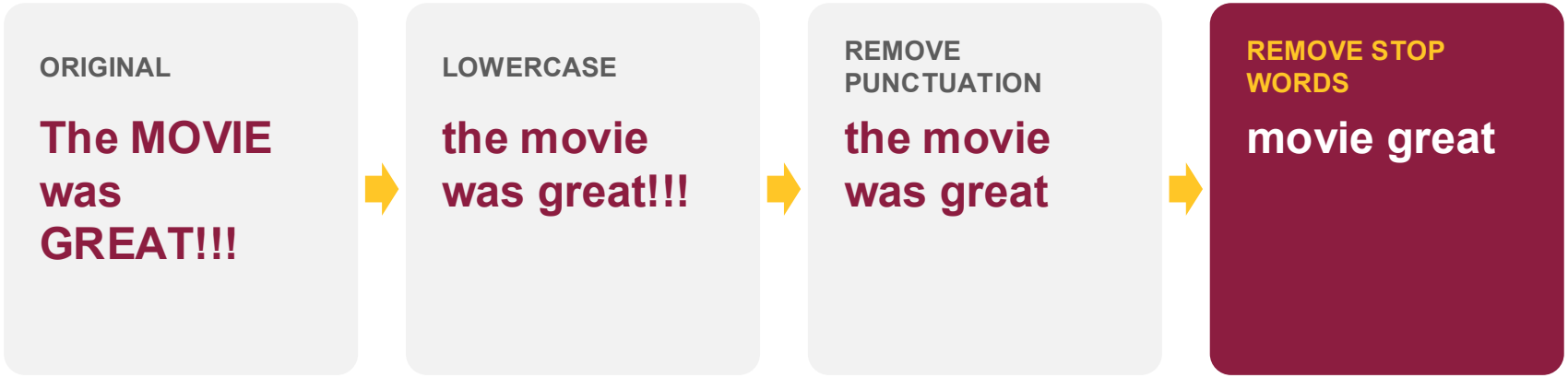


The **movie** **was** **great** **!**

Text Preprocessing

- **Standardizing text**
  - (lowercasing, removing punctuation, handling numbers)
- **Removing “stop words”**
  - (common but uninformative words)
- **Stemming or lemmatization:**
  - Reducing words to their base form
- **Tokenization**
  - Breaking text into smaller units

Text Preprocessing



Stop words are very common words like “the” or “was.”
Cleaning is a choice, not a rule: sometimes those words carry meaning.

Section 2. Numerical Representations



Bag of Words

How many times does a word appear in the corpus?



Document-Term Matrix

How many times is a word present across documents?

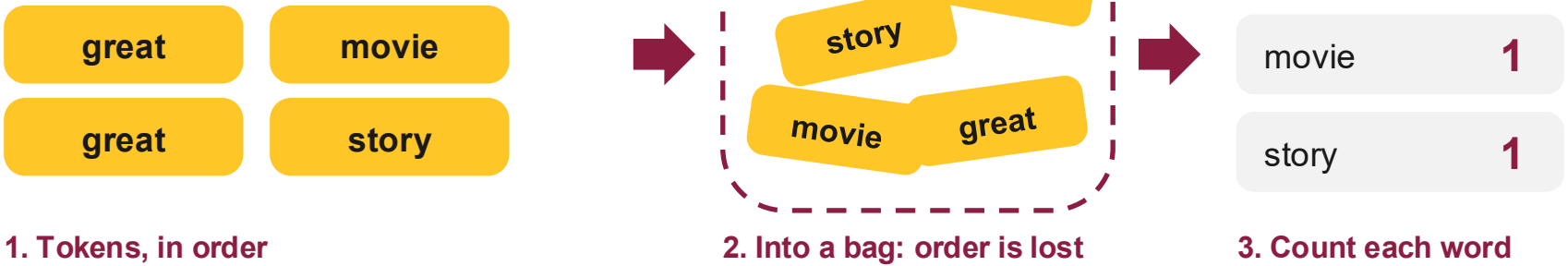


TF-IDF

What words are most influential?

Bag of words

“Great movie! Great story.”



The caveat

Word order and grammar. “Dog bites man” and “man bites dog” get the exact same counts.

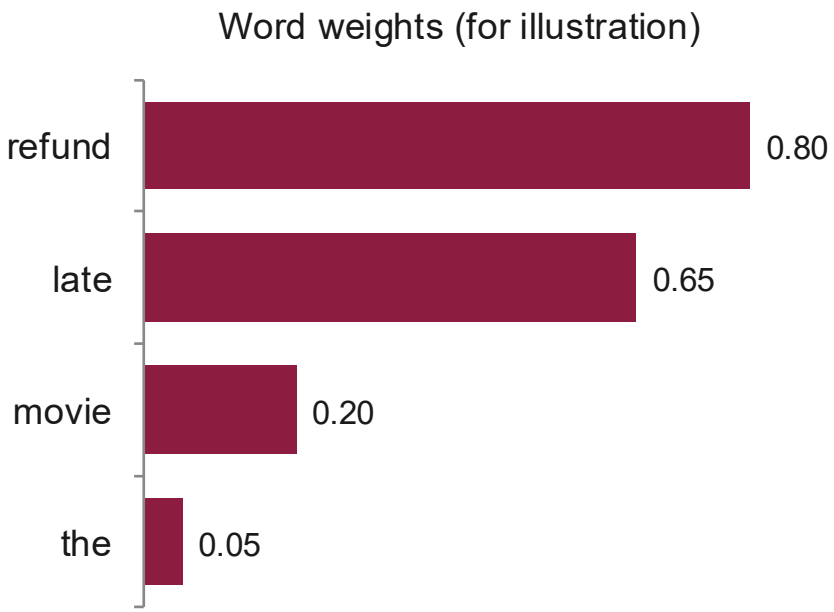
The Document-Term Matrix

D1 The app is helpful
D2 The app crashes often
D3 The update is helpful

In a **Document-Term Matrix**, rows represent documents, columns represent terms, and cells contain numerical values

	app	crashes	helpful	often	the	update
D1	1	0	1	0	1	0
D2	1	1	0	1	1	0
D3	0	0	1	0	1	1

Term Frequency-Inverse Document Frequency



TF-IDF

Gives more weight to distinctive words and less to common ones.

Finding relationships

Documents that share distinctive words end up “close” together.

Section 3. Foundational NLP Applications



ORGANIZING

Text Classification

Which category does this text belong to?



OPINIONS

Sentiment Analysis

Is this text mostly positive or mostly negative?



DISCOVERY

Topic modeling

What are the themes present in a large text?

Text Classification

Message A

From: rewards-center3452@prize-claim-now.biz
Subject: CONGRATULATIONS!!! You have been SELECTED

Dear valued customer,
Claim your FREE $500 Visa gift card within 24 HOURS.
Click here to verify your account.

Message B

From: amerca24@asu.edu
Subject: Final presentation files

Hi team,
Happy Friday! This is a friendly reminder to please send in your final slide deck by 11:59 pm tonight. Thank you!

Text Classification

1 Labeled examples

“Win a free prize now”

spam

“Meeting moved to 3 pm”

not spam

2 Learn patterns

free → spam

prize → spam

meeting → not spam

Each word earns a weight for each label

3 Predict a label

“Claim your free gift”

spam

The word weights add up

Supervised Machine Learning: trains using labeled data, but we hold back some labeled examples it has never seen for testing. The ratio it labels correctly is its accuracy.

Sentiment Analysis

How do you feel when you read these? What emotions do they evoke in you?

Exhibit A

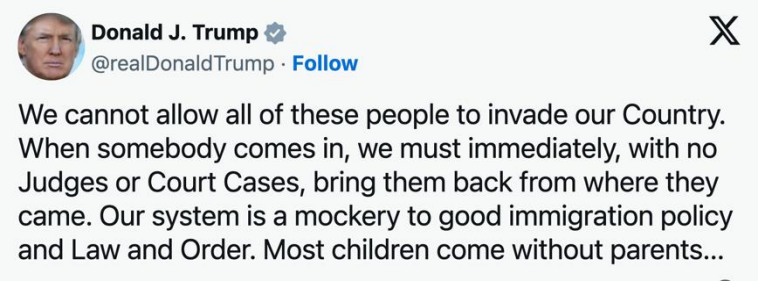


Exhibit B



Sentiment Analysis



How it works

Each word has a score in a dictionary. Rules adjust for “not,” “very,” and “!”, then the scores combine.

Where it struggles

Sarcasm, slang, and words that change meaning by field (“sick” can be praise).

Sentiment Analysis

Table 4. Fixed Effects Regression Predicting Within-User Change in Sentiment Score of Tweets

Variables	Immigrant		Mexican		Hispanic		Asian		Black	
	AZ	NV	AZ	NV	AZ	NV	AZ	NV	AZ	NV
Post–SB 1070 …	.00	.00	−.01**	−.00	−.01	−.02	.01	.00	−.00	.001
	(.00)	(.00)	(.00)	(.00)	(.01)	(.01)	(.01)	(.01)	(.00)	(.0009)
Constant …	−.07***	−.07***	.00*	.01*	.00	.02**	.02***	.03***	.01***	.015***
	(.00)	(.00)	(.00)	(.00)	(.00)	(.00)	(.00)	(.00)	(.00)	(.0004)
Observations …	40,408	5,453	17,543	5,430	3,506	773	3,034	1,588	8,685	3,060
R^2 …	.159	.169	.337	.310	.451	.500	.466	.438	.285	.271

Source: *Do Anti-Immigrant Laws Shape Public Sentiment? A Study of Arizona's SB 1070 Using Twitter Data (2017)*

Topic Modeling

What is the central topic of each message? How can you tell?

Exhibit A

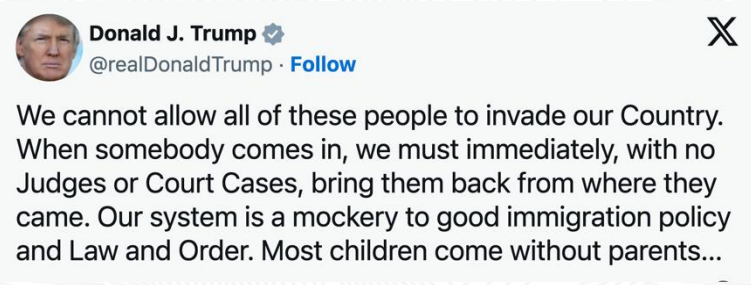


Exhibit B



Topic Modeling

Topic 1

game | team | season

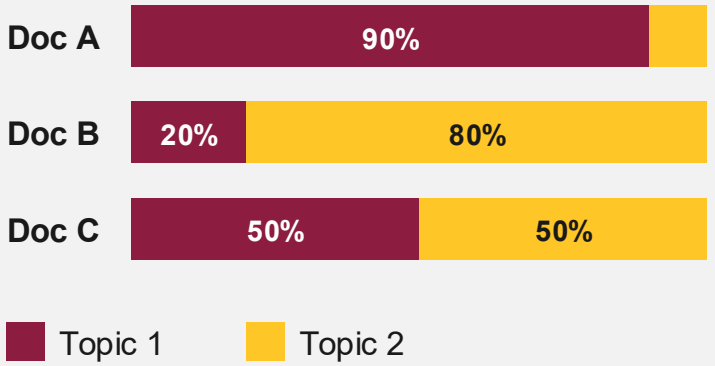
You name it:"Soccer"

Topic 2

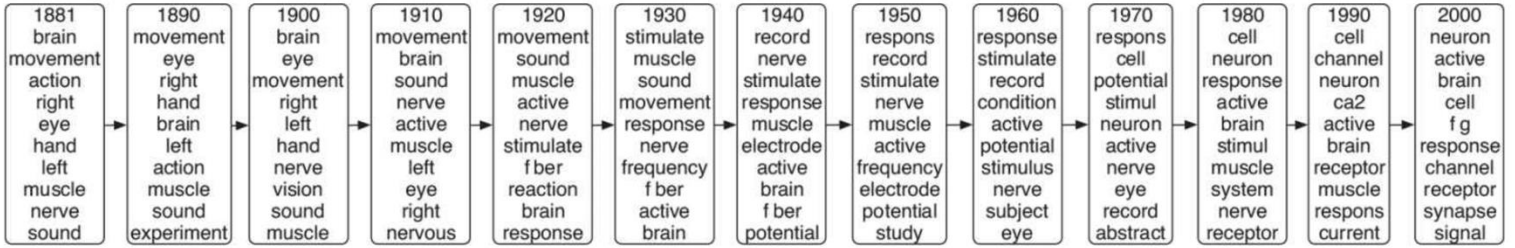
launch | orbit | moon

You name it: "Space"

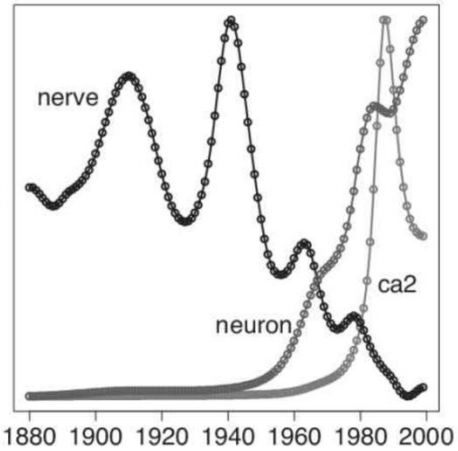
Every document is a mix



Topic Modeling



"Neuroscience"



1887 Mental Science
1900 Hemianopsia in Migraine
1912 A Defence of the ``New Phrenology''
1921 The Synchronal Flashing of Firef ies
1932 Myoesthesis and Imageless Thought
1943 Acetylcholine and the Physiology of the Nervous System
1952 Brain Waves and Unit Discharge in Cerebral Cortex
1963 Errorless Discrimination Learning in the Pigeon
1974 Temporal Summation of Light by a Vertebrate Visual Receptor
1983 Hysteresis in the Force-Calcium Relation in Muscle
1993 GABA-Activated Chloride Channels in Secretory Nerve Endings

Blei and Laferti (2006)

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Demo: Some Practice with Colab

1

Loading our text

Frankenstein, Dracula, Dorian Gray from Project Gutenberg

2

Text to numbers

Clean and tokenize, then TF-IDF distinctive words

3

Sentiment Analysis

VADER scores and the shape of a story

4

Topic modeling

LDA on State of the Union speeches

5

TAKE-HOME

Classification

Which novel is this passage from?

Run the notebook from the top: each part loads its own libraries

NLP in research

Social sciences

Topic modeling · Stance

Immigration tweets around the 2020 election

Hommadova-Lu & Mejova, 2025

Culture and society

Clustering · Topic modeling

Patterns in 2,532 school book bans

Goncalves et al., 2024

Health and medicine

Information extraction

A review of NLP for clinical notes and records

Wang et al., 2018

Engineering education

Topic modeling (LDA)

Topics in student engineering design journals

Romero et al., 2024

Research literature

Topic modeling (LDA)

Themes across 17,393 journal articles

Aslan et al., 2026

Higher education

Sentiment analysis

Sentiment in course evaluation comments

Kocaoğlu & Karataş, 2025

Think about it: what collection of text could you analyze in your own work?

Questions?

Type in the chat or raise your hand

Thank you for joining!



Papers mentioned

1. Hommadova-Lu & Mejova (2025). Immigration Discourse in a Polarized Era: Insights from Social Media Before and After the 2020 US Presidential Election. International Journal of Intercultural Relations, 106, 102162.

2. Goncalves, Langrock, LaViolette & Spoon (2024). Book Bans in Political Context: Evidence from US Schools. PNAS Nexus, 3(6), pgae197.

3. Wang et al. (2018). Clinical Information Extraction Applications: A Literature Review. Journal of Biomedical Informatics, 77, 34–49.

4. Romero, Feijoo-Garcia, Nanda, Newell & Magana (2024). Evaluating the Performance of Topic Modeling Techniques with Human Validation to Support Qualitative Analysis. Big Data and Cognitive Computing, 8(10), 132.

5. Aslan, Aktı Aslan, Turgut, Allison & Özyurt (2026). Thematic Evolution of Educational Technology Research: A Large-Scale Topic Modeling Perspective. Journal of Educational Computing Research, 64(5), 1158–1186.

6. Kocaoğlu & Karataş (2025). Improving Course Evaluation Processes in Higher Education Institutions: A Modular System Approach. PeerJ Computer Science, 11, e3110.